

HEATMAP NODE · REQUIREMENTS

HeatMap Node requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Core above the arm carried into the model (2026-10-02): R7 note, R15 mass 2.08 kg head and 4.92 kg node; no status changes
0.6	2026-10-01	Amish Chadha	Status from HMN-CAL-001 v0.4 after the design for construction (HMN-DDR-003); R13 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3: R13 redefined to the sensor head per HMN-DDR-001 D1; status table replaced by the results of HMN-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node requirements

These are first-pass requirements for the concept. Targets are not yet validated with users and will be revised after co-design sessions (see HMN-PRB-001). Version 0.4 applies the decisions Amish accepted on 2026-09-25 (HMN-DDR-001 and HMN-DDR-002): R4 now covers 0.8 to 20 m/s with calm intervals flagged, R8 now asks for +50 °C at the node, R13 now excludes the FieldNode core and its pole adapter, and R15 now applies to the sensor head only. R1 keeps its ± 0.5 °C target, which the fan-aspirated shield is meant to meet. Version 0.5 raises the R13 target from \$120 to \$130 after Amish approved the budget top-up on 2026-09-26 (HMN-DDR-002 v0.2). Version 0.6 takes its status from HMN-CAL-001 v0.4, after the design for construction (HMN-DDR-003), and reports R13 against `budget_usd`, which Amish confirmed on 2026-10-01 is a hypothetical value-engineering target, not a spending limit. Nothing has been tested.

One requirement is **not met** on paper (R9), three are **at risk** (R5, R10 and R14), eight are met on paper and two are met by design. The sensor head's estimated cost is \$6 over its value-engineering target (R13). See Table 2.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Air temperature accuracy in the street	± 0.5 °C against a reference aspirated thermometer, in full sun at wind speeds of 1 m/s or more; ± 1.0 °C below 1 m/s (met at all wind speeds with the fan-aspirated shield, HMN-DDR-002)	Shield error estimate; later side-by-side field comparison
R2	Relative humidity accuracy	± 3 %RH from 20 to 80 %RH at 15 to 40 °C	Datasheet; CalRig salt fixed points
R3	Globe temperature	Standard 150 mm black globe (emissivity about 0.95); probe ± 0.3 °C after calibration; 90 % response within 30 min	Calibration in CalRig; response estimate
R4	Wind speed	0.8 to 20 m/s; ± 0.5 m/s or ± 10 %, whichever is larger; the share of each interval below the start-up speed is reported so that the server can flag biased intervals (restated under HMN-DDR-002; was 0.5 to 20 m/s)	Datasheet of the chosen anemometer; later wind comparison
R5	Derived heat stress	Mean radiant temperature and outdoor WBGT computed per ISO 7243 definitions with the Liljegren method; WBGT within ± 1.0 °C of a reference WBGT meter	Calculation note; later collocation with a reference meter
R6	Reporting	15 min means of all channels by default (adjustable 5 to 60 min); data at the gateway within 30 min; 7 days stored on the node for resend	FieldNode firmware design; airtime calculation
R7	Power autonomy	No mains; 5 days without sun at the default reporting interval	FieldNode energy budget plus the sensor load in HMN-PRC-001
R8	Outdoor survival	Electronics IP65; operate at -20 to +50 °C at the node (restated under HMN-DDR-002, was +60 °C), with FieldNode's hot-climate sun shield on the core; UV-stable plastics; sensors survive driving rain and dust	Design review; datasheets
R9	Measurement height	Sensors at pedestrian height (1.1 to 2.0 m)	Mounting review with pole owners
R10	Fit to existing poles	Stainless band clamps for 60 to 200 mm round poles; no drilling; fitted from a ladder or lift by two people in 30 min or less	Design review; later timed trial
R11	Wind survival	No failure at a 35 m/s gust; arm deflection under 10 mm at 20 m/s	Wind load calculation
R12	Privacy	Environmental data only: no camera, microphone, Wi-Fi or Bluetooth scanning, or any personal data	Design review of hardware and firmware
R13	Cost	Sensor head (all parts except the FieldNode core and its pole adapter) against a value-engineering target of \$130 (raised from \$120 under HMN-DDR-002 v0.2, budget top-up approved by Amish on 2026-09-26), the <code>budget_usd</code> in <code>project.yaml</code> ; the target is a hypothetical control target, and cost is reported as over or under it. The FieldNode core, its sun shield and the pole	Priced BOM

ID	Requirement	Target	Verification (TRL 3 or later)
		adapter are counted against FieldNode (HMN-DDR-001 D1, HMN-DDR-002); the full-node cost is reported for information	
R14	Traceable data	Each node carries a calibration record from CalRig; readings published in an open, documented format with node location and height	Documentation review
R15	Mass and loading on the pole	4 kg or less for the sensor head (restated under HMN-DDR-002, was the complete node); the complete-node mass is reported for the pole owner; no part larger than 300 mm across other than the arm	Mass estimate

Table 2. Status from HMN-CAL-001 v0.4 (TRL 3 calculations), not met first.

ID	Value (HMN-CAL-001)	Status
R9	Sensors at 2.67 to 2.69 m; pedestrian-height air 0.2 to 0.8 °C warmer in strong sun	Not met
R5	Input error ± 0.30 °C RSS, ± 0.55 °C worst case with the fan; calm intervals flagged; Liljegren model error unknown	At risk
R10	120° V-saddle fits 60 to 200 mm; 37 min estimated	At risk (time)
R14	Probe calibrated before fitting fits a CalRig bay; the assembled 150 mm globe does not; payload format defined	At risk
R1	0.43 °C with the fan at any wind speed; 1.04 °C at 1 m/s with the fan off	Met on paper (fan flow assumed)
R3	Probe ± 0.18 °C (± 0.26 °C with a maximum-tolerance reference); 90 % response about 12 min	Met on paper
R4	Start-up about 0.8 m/s (typical, unverified); calm share reported and flagged at 20 % of an interval	Met on paper (start-up unverified)
R6	20 B payload; 23.7 s/day at SF9; 7 days in 21.5 kB; 16 min latency for 99 % of intervals	Met on paper
R7	Sensors and fan 35.4 mW, 35 % of FieldNode's 100 mW design allowance	Met on paper (the head no longer shades the panel)
R8	FieldNode interior 57.2 °C at 50 °C air with its sun shield, against 70 °C; sensor head parts within ratings	Met on paper (shield factor assumed)
R11	Arm factor 13.0 on yield; 0.31 mm at 20 m/s; clamp twist factor 2.5	Met on paper (preload assumed)
R13	Value-engineering target: USD 130. Estimated cost of the constructable design: USD 136 (USD 6 over the target). FieldNode core with shield and adapter \$156.00; full node \$292.00	Over the target by \$6.00
R15	Sensor head 2.08 kg; complete node 4.92 kg for information; largest part 290 mm	Met on paper
R2	SHT45 ± 1.0 %RH typical	Met by design
R12	No imaging, audio or radio scanning parts; environmental payload only	Met by design

■ Assumptions

- The FieldNode core performs as stated in FND-CAL-001 (100 mW design sensor allowance, 5.75 days without sun at that load; hot-climate node with the sun shield 2.61 kg, \$148.00 after its design for construction, FND-DDR-003).
- WBGT is derived, not measured with a wetted wick, because a natural wet-bulb sensor needs a water reservoir that would need refilling at every node.
- Pole owners permit clamped attachments at the proposed height.